

# Amritpal Singh

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## Skills

- **Languages and Tools:** C, C++, Python, HTML, CSS, Javascript, x86\_64 Assembly, Bash, GDB, CMake, GNU Make, Git, GNU Emacs, VSCode, vim, neovim
- **Other Relevant information:** Custom Linux Kernel, Self Hosted services, IoT, Homelab, Publishing and maintaining a package on AUR.

## Education

I.K.G. Punjab Technical University, B.Tech in Computer Science and Engineering CGPA: 7.4 – April 2027,

## Open-Source Experience

**GNU Emacs (Open Source Contributor)** Commit: 46b6d1

- Resolved a critical decompression vulnerability relating to cross-implementation gzip stream boundaries; engineered a robust verification check to ensure seamless compatibility with alternative implementations like pigz.

**slash\_tmp (Open-Source Contributor)** Pull Request List

- Architected core architectural features including an HTTP chunked transfer encoding protocol, preventing high-concurrency server crashes by reducing memory footprint to  $O(1)$  relative to file size. Implemented native POSIX/Linux signal handling routines to ensure graceful server shutdowns and daemon-like stability. Enhanced network security and protocol compliance by integrating SSL (HTTPS) support and fixing payload mismatches.

**TLP — Laptop Power Management Utility** Commit: 9f3ff4

- Fixed a critical startup initialization failure by refactoring non-POSIX compliant timeouts into standardized, portable alternatives, enabling reliable execution across non-GNU Linux environments.

## Projects

**FORTH86** x86\_64 Assembly, CMake

- Developed a high-performance, Turing-complete FORTH interpreter entirely in bare-metal x86\_64 assembly language. Engineered a custom standard library and wrote low-overhead re-implementations of standard C library (`libc`) routines to optimize execution speed and binary size.

**sinkhole (Custom Operating System Kernel)** C++, C, x86 Assembly, Make

- Designed a monolithic operating system kernel featuring real-time physical memory mapping, dynamic allocation, and custom IDT/GDT initialization.
- Integrated modern security standards by implementing compiler-level stack-smashing protection mechanisms alongside a minimal C++ runtime layer. Configured low-level serial I/O and BIOS VGA buffer for diagnostics.

**perfmode** C, Make

- Built a lightweight Linux CLI system utility interacting directly with ASUS Notebook kernel modules and ACPI platform profiles to regulate hardware fan states and power envelopes. Delivered an early open-source solution providing hardware control for ASUS laptops before official driver software support was introduced, maintaining deep backwards compatibility with legacy third-party drivers.

**silly\_packer** C++, Raylib, CMake, stb

- Developed an optimized texture packing and asset serialization asset pipeline utility that compiles modular graphics into compressed, game-ready texture atlases. Programmed an automated C++ header generator supporting static binary blob embedding and optional API wrappers for fluid integration into production game loops.

**c8fontgen** C, Raygui, Raylib, CMake

- Created a graphical pixel-editor utility optimized for designing custom bitmap fonts for CHIP-8 emulators, providing seamless clipboard integration and structured hexadecimal output parsing.

**Infrastructure & Homelab DevOps** Docker, Linux, Networking

- Provisioned and maintained a self-hosted cloud infrastructure platform leveraging bare-metal container runtimes (Docker/Podman) and secure network topologies.
- Configured encrypted reverse-proxy routing via remote FRPS tunnels traversing Oracle Cloud infrastructure to securely expose decentralized communication endpoints (Matrix Homeserver).
- Deployed high-availability local services including centralized media delivery systems (Jellyfin, \*arr stack) and network-level security via BlockyDNS routing.